

Express-Kart is a powerful, multi-seller E-commerce platform inspired by premium shopping experiences. Built with Python and Flask, it offers a robust set of features for both customers and merchants.

1. Core Technologies

- **Backend:** Python 3.x, Flask (Web Framework)
- **Database:** SQLite (Serverless, production-ready)
- **Frontend:** HTML5, Vanilla CSS3, Javascript (Vanilla)
- **Authentication:** bcrypt (Password hashing), OTP-based Verification
- **Payments:** Razorpay Integration
- **Email:** Flask-Mail (SMTP Integration)
- **Utilities:** xhtml2pdf (Invoicing & Documentation)

2. Project Folder Structure

```
Ecommers/
■■■■ app.py                # Main application logic & routes
■■■■ auth_utils.py         # OTP generation & verification helpers
■■■■ config.py             # Application configurations (DB, Mail, API keys)
■■■■ ecommerce.db          # SQLite Database file
■■■■ forgot_password.py    # Password recovery module
■■■■ init_db.py            # Database schema initialization script
■■■■ static/              # Static assets
■   ■■■■ uploads/         # User/Merchant uploaded images
■   ■■■■ style.css        # Main stylesheet
■   ■■■■ auth.css         # Authentication specific styles
■   ■■■■ script.js        # Main client-side logic
■■■■ templates/           # Jinja2 HTML templates
■   ■■■■ base.html        # Global layout template
■   ■■■■ navbar.html      # Customer navigation
■   ■■■■ merchant_navbar.html # Merchant portal navigation
■   ■■■■ auth/            # Auth-related templates (OTP, Reset)
■   ■■■■ merchant/        # Merchant dashboard & product management
■   ■■■■ user/            # Customer views (Cart, Orders, Products)
■■■■ utils/               # Extension modules
    ■■■■ pdf_generator.py  # PDF generation utility
```

3. Database Overview

The system uses a relational SQLite database with the following core tables:

Table	Purpose
users	Stores customer profiles and login credentials.
admin	Stores merchant/admin accounts with role-based access.

products	Inventory management, linked to specific merchants.
orders	Payment tracking and high-level order information.
order_items	Detailed breakdown of products within each order.

4. Key Module Descriptions

app.py

The heartbeat of the application. It handles routing for product browsing, cart management, checkout flows, and merchants' item management. It also contains the Razorpay payment verification logic.

config.py

Centralizes all sensitive information and app settings. Allows for easy configuration of database paths, email server credentials, and Razorpay API keys.

forgot_password.py

A dedicated blueprint for password recovery. It coordinates the OTP sending process and updates credentials securely for both users and admins.

5. Features Highlight

- **Multi-Seller Support:** Multiple merchants can sign up and manage their own product catalogs.
- **Secure Authentication:** Industry-standard password hashing using bcrypt.
- **OTP System:** Email-based OTP for registration and password resets.
- **Dynamic Cart:** Session-persistent shopping cart with seamless add/remove.
- **Razorpay Integration:** Secure online payment processing.
- **Detailed Invoicing:** PDF invoices generated automatically upon successful purchase.
- **Loading Spinner:** Modern UX feedback for background processes like email delivery.